

```

epochs, number of samples
learning_rate = 0.01
epochs=100
n_samples = 30

train_x = np.linspace(0, 20, n_samples)
train_y = 3 * train_x + 4 * np.random.randn(n_samples)
## Plot before starting the training
plt.plot(train_x, train_y, 'x')
plt.plot(train_x, 3 * train_x)
plt.show()

# Initialize training weights in placeholder
X = tf.placeholder(tf.float32)
Y = tf.placeholder(tf.float32)

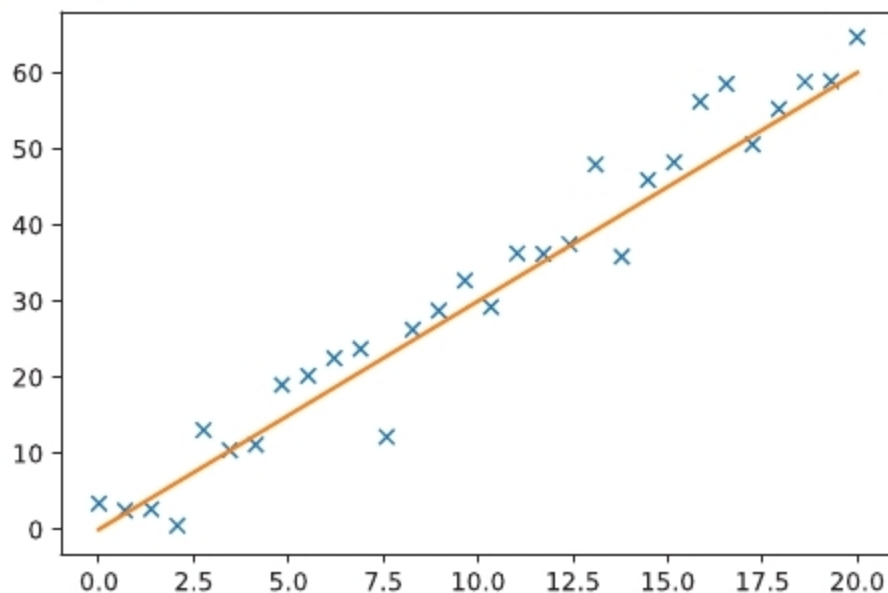
# Now initialize training biases
W = tf.Variable(np.random.randn(), name='weights')
B = tf.Variable(np.random.randn(), name='bias')
# The prediction can be based input X, Weight W and bias B
prediction = tf.add(tf.multiply(X, W), B)

# Cost function(To minimize the loss as much as possible)
# Use mean square error

cost = tf.reduce_sum((prediction - Y) ** 2) / (2 * n_samples)

optimizer = tf.train.GradientDescentOptimizer(learning_rate).
minimize(cost)

```



```
#Global initializer for Tensorflow variables
```

```

init = tf.global_variables_initializer()
# tf.session for all the computation
# here the no.of epochs are run

```

```

with tf.Session() as sesh:
    sesh.run(init)

```

```

for epoch in range(epochs):
    for x, y in zip(train_x, train_y):

```

```

        sesh.run(optimizer, feed_dict = {X: x, Y:y})

    if not epoch % 20:
        c = sesh.run(cost, feed_dict={X:x, Y:y})
        w = sesh.run(W)
        b = sesh.run(B)

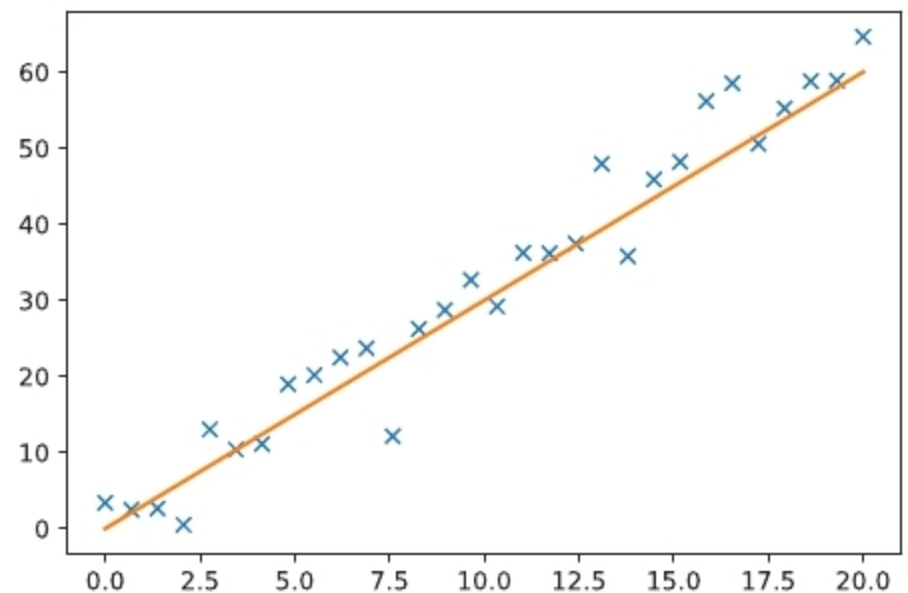
        print(f'epoch: {epoch:04d} cc={c:.4f} w={w:.4f}
b={b:.4f}')
        weight = sesh.run(W)
        bias = sesh.run(B)
        plt.plot(train_x, train_y, 'o')
        plt.plot(train_x, weight * train_x + bias)
        plt.show()

```

```

epoch: 0000 cc=5.4618 w=2.3091 b=0.3990
epoch: 0020 cc=0.0384 w=3.1360 b=0.4447
epoch: 0040 cc=0.0381 w=3.1375 b=0.4217
epoch: 0060 cc=0.0378 w=3.1389 b=0.3999
epoch: 0080 cc=0.0375 w=3.1402 b=0.3792

```



To get familiar with the basics of machine learning and deep learning, go to <https://opensourceforu.com/2017/12/deep-learning-using-algorithms-to-make-machines-think/>. **END** 🐧

References

- [1] <https://medium.com/@jrodthoughts/five-deep-learning-frameworks-that-you-should-know-about-and-some-data-to-back-it-up-4480d31c86a8>
- [2] <https://tensorflow.org/>
- [3] <https://opensourceforu.com/2017/12/deep-learning-using-algorithms-to-make-machines-think/>
- [4] <https://towardsdatascience.com/what-happened-at-the-tensorflow-dev-summit-2017-part-1-3-community-applications-77fb5ce03c52>

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